

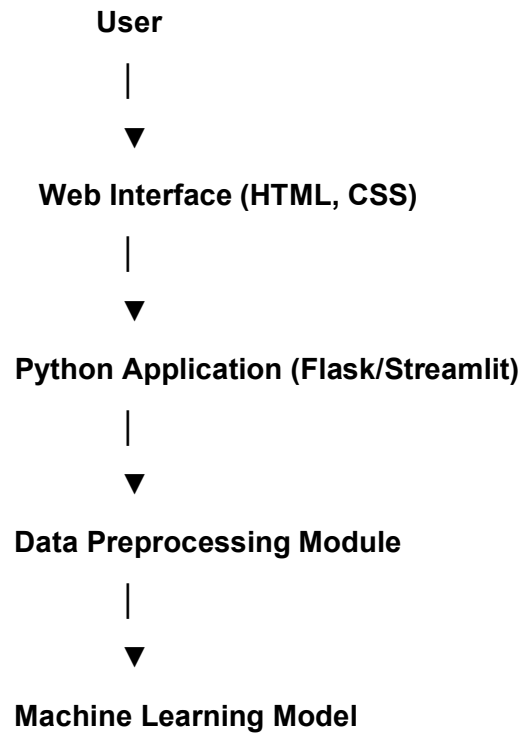
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 July 2026
Team ID	SWTID-2026-2149
Project Name	Credit card approval prediction
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example:



(Random Forest / Logistic Regression)



Prediction Result

(Approved / Rejected)



Dataset / Database

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	User enters applicant details and views prediction result.	HTML, CSS, Streamlit
2.	Application Logic-1	Process user input and validate data.	Python
3.	Application Logic-2	Data preprocessing (cleaning and encoding).	Pandas, NumPy
4.	Application Logic-3	Generate approval prediction using ML model.	Scikit-learn
5.	Database	Store applicant dataset (if required).	MySQL / CSV Dataset
6.	Cloud Database	Cloud storage (optional).	Not Used
7.	File Storage	Store trained ML model and dataset.	Local File System (.csv, .pkl)
8.	External API-1	Not required	-
9.	External API-2	Not required	-

10.	Machine Learning Model	Predict credit card approval or rejection	Random Forest / Logistic Regression
11.	Infrastructure (Server / Cloud)	Run application on local system.	Local System / Streamlit

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks and libraries used for development.	Python, Pandas, NumPy, Scikit-learn, Streamlit
2.	Security Implementations	Validate user input and protect application data.	Input Validation, Password Protection (if login is used)
3.	Scalable Architecture	The application can be expanded to support more users and larger datasets.	Python, Streamlit
4.	Availability	The application can run whenever the local system or server is available.	Local System
5.	Performance	The ML model provides predictions within a few seconds.	Scikit-learn, Pandas

